MSC DISSERTATION VIVA — MODULE M598

An Uncertainty-Aware, Explainable and Human-Reviewed Framework for Remaining-Useful-Life-Based Maintenance Prioritisation

Evidence from the NASA C-MAPSS FD001 Turbofan Dataset

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Applied quantitative research · Secondary simulated data · Reproducible demonstration

A point RUL prediction hides what the reviewer needs most

Three gaps in a bare number:

No uncertainty — the model says 45 cycles, but its error could be ±108

No explanation — which sensors drove that estimate?

No structured review — no audit trail if a human overrides

The age-only near-tie on validation:

Random Forest	**29.479**
Naive age-only	**29.556**

Δ = 0.077 cycles mean engine RMSE
Within 1% practical-equivalence band
PHM tie-break selected Random Forest

20 validation engines · 4,217 rows

Prior work addresses each piece — none integrates all six

Gap 1: Leaky splits

Row-random partitions leak future sensor values across engine trajectories (Kaufman et al., 2012)

Gap 2: Isolated treatments

Uncertainty (conformal) and explanation (SHAP) studied separately, rarely with a decision layer

Gap 3: No audit

Accountability discussed conceptually but human review rarely implemented and audited

This study's integrated contribution: engine-safe selection + naive baseline + whole-engine conformal intervals + verified SHAP + deterministic triage + separately tested HITL prototype with audit

Key references: Saxena et al. (2008); Jardine, Lin & Banjevic (2006); Lei et al. (2018); Angelopoulos & Bates (2023); Lundberg & Lee (2017); Parasuraman, Sheridan & Wickens (2000); Shneiderman (2020)

FD001: one fault mode, engine-safe partitioning, one official test

Engine-level partition (seed 42, no reroll):

Role	Engines	Rows
Training (modelling)	60	12,310
Validation (selection)	20	4,217
Calibration (conformal)	20	4,104
Official test	100 endpoints	sealed

Each engine has exactly one role → no leakage.
Test opened once after all policies frozen.

Target construction:

$$RUL(i, t) = T_i - t$$

Primary: uncapped · Sensitivity: capped at 130

Dataset characteristics:

- → Single operating condition (sea level)
- → Single fault mode (HPC degradation)
- → Simulated, not real fleet data
- → 14 selected sensors + 3 operational + age

Random Forest won a near-tie via the predeclared PHM penalty

Validation comparison (20 engines, 4,217 rows):

Model	Engine RMSE	Mean PHM penalty
Naive age-only	29.556	156.44
Ridge	31.516	157.53
Random Forest	**29.479**	**83.67**
XGBoost	31.156	101.15

Δ = 0.077 cycles — inside 1% equivalence band.
The frozen PHM tie-break, not RMSE, determined selection.

Selection policy (frozen before test):

Step 1: If any model's mean engine RMSE is outside 1% of the best → eliminate.

Step 2: Among survivors, pick the lowest asymmetric PHM score.

Rationale: PHM penalises optimistic (late) predictions more steeply — exp(e/10) vs exp(−e/13) — aligning with safety preference.

5-fold grouped CV on 60 training engines · No retuning after selection

Thesis §4.7, §5.2 · Tables 4.3, 5.1 · **Backup: Model configs ↗**

Engine-block conformal: valid coverage at a scheduling cost

Construction:

1. For each calibration engine, compute
$s_i = \max_t |\hat{y}(i,t) - y(i,t)|$

2. Rank 20 engine scores

3. Finite-sample rank at 90% → 19th of 20 scores

4. $\hat{q} = 107.90$ cycles

5. Interval: $[\hat{y} - \hat{q}, \hat{y} + \hat{q}]$, clipped at 0

Result on 100 test endpoints:

100/100
OBSERVED ENDPOINT COVERAGE

182.46
MEAN INTERVAL WIDTH (CYCLES)

63%
LOWER BOUNDS CLIPPED TO ZERO

Wilson 95% CI for coverage: [96.3%, 100%]
20 calibration engines · Nominal 90%

Thesis §4.8, §5.4 · Angelopoulos & Bates (2023) · **Backup: Conformal rank ↗**

Official test: useful accuracy with systematic optimistic bias

18.88	25.73	+10.90	107.90
MAE (CYCLES)	RMSE (CYCLES)	MEAN BIAS (OPTIMISTIC)	CONFORMAL Q̂ (CYCLES)

Post-closure naive comparison:

	MAE	RMSE	Bias
Random Forest	18.88	25.73	+10.90
Naive (post-closure)	30.88	—	—

⚠ Naive endpoint MAE is a post-closure reference — reuses sealed outcomes

Bias by true-RUL group (Table 5.4):

Group	n	MAE	Bias
CRIT 0–30	25	10.52	+8.97
NEAR 31–60	14	7.34	+3.95
MID 61–130	53	26.49	**+15.20**
EARLY >130	8	14.82	+0.53

Bootstrap 95% CI (5,000 engine resamples): MAE 15.56–22.48 · RMSE 21.33–30.15 · bias 6.47–15.57

Ablations expose sensitivity: age dominates; narrower intervals lose trajectory coverage

Feature ablation (post-closure, validation):

Configuration	Pooled RMSE	Δ
All features (baseline)	33.07	—
Remove age	41.16	+24.5%
Remove sensors 9/14	31.76	−4.0%

60 modelling engines · 20 validation engines
Fixed RF configuration, no retuning

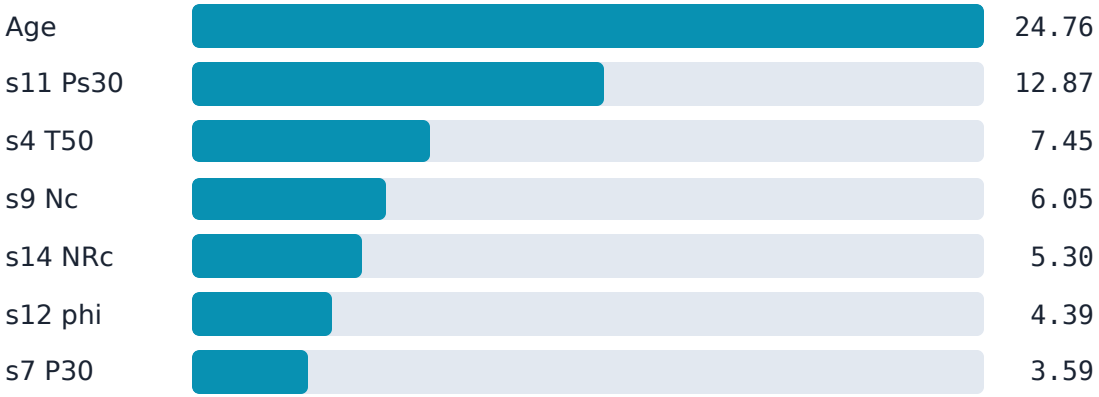
Conformal score comparison (post-closure):

Score	Width	End. Cov.	Traj.
Engine-max	182.46	**100/100**	**20/20**
Row-level	106.34	96/100	8/20
Engine-mean	100.72	96/100	8/20

Trade-off: 42–45% narrower intervals ↔ only 8/20 validation trajectories fully covered

SHAP audits predictions — age and HPC sensors dominate, but cannot prove causation

Mean |SHAP| at 100 official endpoints (cycles, top 7 of 18):



Baseline prediction: 107.84 cycles
Additivity verified: Σ SHAP ≈ prediction

Failure case — Engine 41 (truly CRITICAL, point class MEDIUM):

Age 123 · **true RUL 18** · **predicted 60.0**

`baseline 107.84`

`age −29.2 · T50 (s4) −11.5 · phi (s12) −7.9`

`Nc core speed (s9) +4.3 — partly counteracts`

Engine 18 shows the same pattern (true 28, predicted 64.5)

SHAP supports numerical reconstruction — it does not establish physical causation. Correlated sensors can exchange credit.

Thesis §5.8 · Tables 5.8, 5.9 · Lundberg & Lee (2017)

The offline guard catches all critical engines — but reviews 81 of 100

Point-only triage (no interval):

62/100
CORRECTLY CLASSIFIED

34/100
UNDER-TRIAGED

9 of 25 truly CRITICAL engines were **not** labelled CRITICAL by point prediction alone

With offline lower-bound guard (post-closure, not in n8n):

39/39
CRITICAL+HIGH CAUGHT

81/100
REFERRED TO REVIEW

47 of 81 entered on interval condition alone.
Trade-off: 100% critical recall ↔ 81% review burden.

Validation rows: 2,578/4,217 referred · 2/1,220 truly critical-or-high rows missed · These are repeated measurements, not independent engines.

Separately tested n8n prototype: fixed payload, human authority, audit trail

TRACK 1 — EVALUATED OFFLINE ANALYTICAL PIPELINE

FD001 Data → Engine Split → Random Forest → Conformal Interval → SHAP → Offline Triage Guard

⚠ No implemented live connection — proposed future link

TRACK 2 — SEPARATELY TESTED FIXED-PAYLOAD N8N PROTOTYPE (22 NODES)

Fixed Payload → Policy Engine → Bounded LLM → Human Review → Audit DB

Audit evidence:

Decision	Records	Status
APPROVE	2	Exported
REJECT	1	Exported
MODIFY (valid)	1	Exported
MODIFY (invalid)	—	Route trace only

Key boundaries:

- → Fixed illustrative payload, no live feed
- → Offline guard NOT in n8n routing
- → LLM narrates under constraints, does not decide
- → Deterministic policy set before narration
- → No production readiness or certification claim

Thesis §4.10, §5.10 · Tables 5.11, A.1 · Figure 4.6 · **Backup: Prototype detail ↗**

Three evidence-supported conclusions — and what comes next

1

Rigorous evaluation design

Engine-safe RF improved on age under PHM penalty. The near-tie in RMSE is an honest finding.

2

Coverage at a scheduling cost

100/100 endpoint coverage with 182-cycle width. Narrower alternatives lose trajectory validity.

3

Implementable human review

39/39 critical recall at 81% review burden. Audit trail demonstrates accountability.

Limitations:

- → Single simulated fault mode (FD001)
- → Conformal width too broad for scheduling
- → No live data feed; fixed prototype payload
- → Provenance discontinuities in legacy hashes

Future work:

- → FD002–FD004 and N-CMAPSS evaluation
- → Narrower whole-engine-valid intervals
- → Live prediction service with authentication
- → Usability study with maintenance engineers

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Repository: github.com/mranitabh/GISMA-hitl-dissertation-predictive-maintenance

Research questions → Evidence mapping

RQ	Question	Evidence
RQ1	How accurately can lightweight ML estimate RUL under an engine-level, leakage-controlled design vs a naive age-only baseline; which feature groups drive accuracy?	MAE 18.88, RMSE 25.73, bias +10.90; PHM tie-break selected RF (§5.2–5.3, §5.6)
RQ2	How can engine-block conformal prediction give intervals with a transparent finite-sample rule; what coverage–precision trade-off results?	q̂ 107.90; 100/100 endpoints; width 182.46; alternatives 8/20 trajectories (§5.4, §5.7)
RQ3	What do SHAP explanations reveal, read against turbofan physics without causal claims?	Additive reconstruction; age dominant; core-speed counter-contributions (§5.8)
RQ4	How can prediction, uncertainty and explanation become CRITICAL/HIGH/MEDIUM/LOW triage with mandatory review — how safe and burdensome?	Point under-triage 34/100; guard 39/39 at 81/100 reviews; 3 decision types recorded (§5.9–5.10)

Engine-level split prevents trajectory leakage

Why row-random splits leak:

In a turbofan trajectory, consecutive cycles share autocorrelated sensor readings. A row-random split places cycle t in training and cycle t+1 in test — the model effectively sees the answer. Engine-level splitting ensures no cycle from a test engine ever appears in training.

> Fixed seed 42 · No rerolling · No stratification
> Each of 100 training engines assigned to exactly one role

Partition details:

Role	Engines	Rows	Purpose
Modelling	60	12,310	Train models
Validation	20	4,217	Select model
Calibration	20	4,104	Set $\hat{q}$
Official test	100	endpoints	One-time eval

Reference: Kaufman et al. (2012) on leakage

Model configurations and selection criteria

Model	Key Parameters	CV	Notes
Naive age-only	$\hat{y} = \max(195 - \text{time_in_cycles}, 0)$	Median training lifetime	No sensors used
Ridge	$\alpha = 0.0001$ (grid boundary)	11 α values, grouped 5-fold	Scaler fit on training engines only
Random Forest	depth 8, min leaf 10, max_features 0.5, 500 trees	18 candidates, grouped 5-fold	Raw inputs
XGBoost	depth 3, learning rate 0.03, …	18 candidates, grouped 5-fold	Raw inputs

All configurations frozen before validation evaluation. Outputs clipped to ≥ 0. No retuning after model selection.

Conformal score: rank calculation and assumptions

Score computation:

For calibration engine i:

$s_i = \max_{t \in \text{trajectory}} |\hat{y}(i,t) - y(i,t)|$

Sort 20 scores; finite-sample rank for 90% nominal coverage:

19th of 20 → $\hat{q} = 107.8965 \approx 107.90$ cycles

Calibration covered 19/20 engines by construction — not validation

Key assumptions:

- → Exchangeability of calibration and future engines
- → Coverage is marginal, not conditional
- → Not guaranteed per-engine or per-trajectory
- → Calibration is separate from fitting

Why engine-maximum?

Row-level scores give valid row-marginal coverage but can miss the worst-case point in a trajectory. The engine-maximum ensures the interval would have covered every observed point for at least 90% of calibration engines.

The cost: the worst row drives $\hat{q}$ for the entire engine, producing wide intervals (182.46 cycles mean width).

References: Vovk, Gammerman & Shafer (2005)
Angelopoulos & Bates (2023)
Lei et al. (2018a)

Triage policy definitions and metric denominators

Lifecycle thresholds (stipulated):

Class	Rule 1 on predicted ŷ	Review
CRITICAL	ŷ ≤ 30	Mandatory
HIGH	30 < ŷ ≤ 60	Mandatory
MEDIUM	60 < ŷ ≤ 130	Routine unless Rule 2
LOW	ŷ > 130	Routine unless Rule 2
Rule 2 (offline, post-closure): refer if conformal lower bound ≤ 30		

⚠ These are research assumptions, not maintenance-approved limits

Metric definitions:

Critical-class recall:
correctly CRITICAL / all truly CRITICAL

CRIT+HIGH review recall:
referred CRIT+HIGH / all truly CRIT+HIGH = 39/39

Review burden:
total referred / total endpoints = 81/100

Validation rows: 2,578/4,217 referred
Repeated rows per engine — not independent

n8n prototype: 22-node topology and audit record

Workflow stages:

1. **Input:** PROTOTYPE_TEST fixed payload
2. **Validation:** Model provenance check
3. **Policy:** DEMO_POLICY_V1 + STIPULATED_RESEARCH_ASSUMPTION
4. **Narration:** local Qwen3 model under constraints
5. **Review form:** APPROVE / REJECT / MODIFY
6. **Persistence:** PostgreSQL audit record

Policy and priority set before LLM narration.
LLM constrained to narrate evidence, not decide.

Audit record fields (representative):

```
Request: request id, asset id, predicted RUL
Policy: DEMO_POLICY_V1 · STIPULATED_RESEARCH_ASSUMPTION
Explanation: SHAP drivers + constrained agent summary
Human decision: decision, comment, revised recommendation
Timestamps: creation, review, completion
```

Repository: github.com/mranitabh/
GISMA-hitl-dissertation-predictive-maintenance

Standards and frameworks — design references, not compliance claims

Standard	Role in Design
ISO 13374	Separates signal processing, health assessment, prognosis, advice
ISO 13381-1	Frames uncertainty reporting requirements
ISO 17359	Informs alert criteria and condition monitoring
ISO 55000	Asset management vocabulary and principles
EN 13306	Maintenance terminology
NIST AI RMF	AI risk management framework
EU AI Act (2024)	Harmonised AI rules — human oversight requirements
IEEE 1856	PHM framework for electronic systems

> Standards inform design decisions — the framework does not claim formal compliance with any standard.

Limitations and their consequences

Limitation	Consequence
FD001: single fault, single regime, simulated	No claim of generalisability to real engines or multi-fault scenarios
Engine-max conformal width (182.46 cycles)	Too broad for precise scheduling; needs narrower valid alternatives
Optimistic bias (+10.90 cycles)	Systematic overestimation — the dangerous direction for maintenance
Fixed prototype payload	No live sensor feed, no live model integration
Offline guard not in n8n routing	Interval-based referral evaluated offline only
4 audit records, manually seeded	Demonstrates architecture, not statistical reliability
Legacy hash discontinuities (Phase 5–6)	Numerical results reproduced but not every earlier artefact's provenance
SHAP correlation sensitivity	Correlated sensors exchange credit; no causal claims
No real technician study	No usability, trust or decision-quality evidence